

# Adam Saidane

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Software Engineering student at INSAT with hands-on experience in backend systems, full-stack development, and applied AI. Built scalable web applications and machine learning systems, including reinforcement learning solutions and production-grade APIs. Seeking a software engineering internship.

## EDUCATION

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| <b>National Institute of Applied Sciences and Technology (INSAT)</b> – Tunisia | Since September 2023       |
| <i>Software Engineering – Ranking: 37/377</i>                                  |                            |
| <b>Menzah 8 Pioneer High School (LPM8)</b> – Tunisia                           | September 2019 – June 2023 |
| <i>Baccalaureate, Mathematics Section - Highest Honors - 18.21/20</i>          |                            |

## WORK EXPERIENCE

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| <b>Teamwill</b> - Tunis, Tunisia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | July 2026 - August 2026 |
| <i>Data Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |
| <ul style="list-style-type: none"><li>Designed and built a generic Talend/Java ETL pipeline synchronizing two unknown PostgreSQL schemas, dynamically discovering tables, columns, and constraints runtime.</li><li>Engineered a streaming row-comparison engine (checksum-based diffing with server-side cursors) achieving O(1) memory per table, avoiding full table loads on datasets of arbitrary size.</li><li>Implemented automatic foreign-key dependency resolution (topological sort) to generate transaction-safe INSERT/UPDATE/DELETE scripts respecting referential integrity constraints.</li><li>Added fault-tolerance layer with retry/backoff logic, connection recovery, and REPEATABLE READ transaction snapshots to guarantee consistent reads under concurrent writes.</li></ul> |                         |
| Stack: Java, Talend Studio, PostgreSQL, JDBC, SQL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |

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| <b>Welyne</b> - Tunis, Tunisia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | April 2026 - June 2026 |
| <i>AI Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                        |
| <ul style="list-style-type: none"><li>Built a FastAPI backend automating monthly financial reporting: CSV ingestion, KPI computation, and AI-generated analysis.</li><li>Designed an asynchronous background pipeline (upload → processing → PDF generation) with real-time task status tracking via REST endpoints.</li><li>Generated automated PDF reports combining financial KPIs and AI-written insights using ReportLab, Pandas, and Matplotlib.</li><li>Wrote unit tests with pytest and pytest-cov to validate data processing and KPI calculations.</li></ul> |                        |
| Stack: Python, FastAPI, Pandas, ReportLab, Pytest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |

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| <b>International Information Development (IID)</b> - Tunis, Tunisia                                                                                                                                                                                                                                                                                                                                                                                                       | July 2025 - August 2025 |
| <i>Software Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |
| <ul style="list-style-type: none"><li>Built Spring Boot REST APIs for banking systems (credit card lifecycle, e-commerce payments) used in production environments.</li><li>Increased API reliability by achieving 96%+ test coverage using JUnit &amp; Mockito, ensuring robust edge-case handling.</li><li>Led 'Edeal' CRM migration, resolving any post-deployment issues.</li><li>Collaborated with engineers to deliver production-ready backend services.</li></ul> |                         |
| Stack: Java, Spring Boot, Swagger, JUnit, Mockito, REST APIs                                                                                                                                                                                                                                                                                                                                                                                                              |                         |

## PROJECTS

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| <b>Open-Source Contributions - Keras</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |
| <ul style="list-style-type: none"><li>Contributed to <b>Keras</b> by implementing input validation for PReLU and LeakyReLU activation layers (PR #22162).</li><li>Diagnosed backend-specific failure in <b>OpenVINO</b> during testing, isolating it as a pre-existing issue.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                           |                                           |
| <b>GitReviewAI</b> – GitHub Code Review Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Python, FastAPI, React, PostgreSQL</i> |
| <ul style="list-style-type: none"><li>Built an AI-powered GitHub code review platform using a <b>multi-agent pipeline</b> (5 specialized LLM agents running in parallel via Celery) that automatically analyzes Pull Requests for security vulnerabilities, performance issues, architecture violations, maintainability, and test coverage.</li><li>Designed a <b>RAG system</b> using tree-sitter code chunking, OpenAI embeddings, and Qdrant vector store to enrich agent prompts with semantic codebase context.</li><li>Implemented production-grade infrastructure: Prometheus metrics, Grafana dashboards, Redis caching, rate limiting, OWASP security headers, and Docker multi-stage builds.</li></ul> |                                           |
| <b>MedAssistAI</b> – Intelligent Medical Assistance Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Python, FastAPI, React, PostgreSQL</i> |
| <ul style="list-style-type: none"><li>Built a full-stack medical platform with an LLM-powered chatbot for symptom collection and AI-driven diagnostic analysis via WebSocket streaming for patients and image analysis module for doctors.</li><li>Designed a Bayesian diagnostic engine scoring <b>500+</b> pathologies across age brackets and symptom profiles.</li><li>Implemented a 5-specialization medical image analysis pipeline reducing analysis time to under <b>60s</b>.</li><li>Secured data with AES-256 file encryption, JWT auth, rate limiting, audit logging, and full GDPR compliance.</li></ul>                                                                                              |                                           |
| <b>AI Traffic Lights Manager</b> – Deep Reinforcement Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Python, PyTorch, SUMO/TraCI</i>        |
| <ul style="list-style-type: none"><li>Built and trained RL agents (Q-Learning, DQN, PPO) for traffic optimization using SUMO simulation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                           |

- Reduced average vehicle waiting time by ~**60%** (1692s → 680s) vs fixed-timing baseline.
- Designed environment with 16 lanes & real-time state tracking, improving policy generalization.
- Benchmarked multiple algorithms and identified DQN as best-performing model.

#### **xG Model** – Football Goal Prediction with Machine Learning

*Python, Streamlit*

- Trained ML models on **10k+** football shots dataset for goal probability prediction.
- Engineered **10+** spatial and contextual features to improve model accuracy achieving **96%** accuracy.
- Built interactive Streamlit app for real-time xG predictions.

### CERTIFICATIONS

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| <b>IBM AI Engineering Professional Certificate</b> · IBM             | 2026 |
| <b>Generative AI Engineering with LLMs Specialization</b> · IBM      | 2026 |
| <b>CS50x — Introduction to Computer Science</b> · Harvard University | 2024 |

### VOLUNTEERING

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| <b>Securinets INSAT</b>                                                                                                                                                                     | Since September 2025       |
| • Active member engaged in technical workshops on reverse engineering and cybersecurity conferences, enhancing practical and theoretical expertise.                                         |                            |
| <b>Institute of Electrical and Electronics Engineers (IEEE)</b>                                                                                                                             | Since September 2023       |
| • Active member involved in programming, algorithms, and AI workshops; competed in multiple competitive programming contests, developing strong analytical and problem-solving abilities.   |                            |
| <b>Tunisian International Model United Nations (TIMUN)</b>                                                                                                                                  | September 2020 – July 2022 |
| • Represented states in UN simulations; prepared resolutions and negotiated on global issues. Developed public speaking, leadership, and formal writing skills in an international context. |                            |

### SKILLS

- **Soft Skills:** Time management - Critical thinking - Team spirit - Sense of responsibility
- **Programming Languages:** Python - Java - C# - PHP - SQL - C / C++
- **Backend:** Spring Boot - ASP.NET Core 8 - Symfony 6
- **ML / AI:** PyTorch - TensorFlow - Keras - Scikit-learn - Pandas - NumPy
- **Tools:** Git - Docker - Postman - Swagger - Jupyter - TensorBoard
- **Languages spoken:** Arabic - English - French